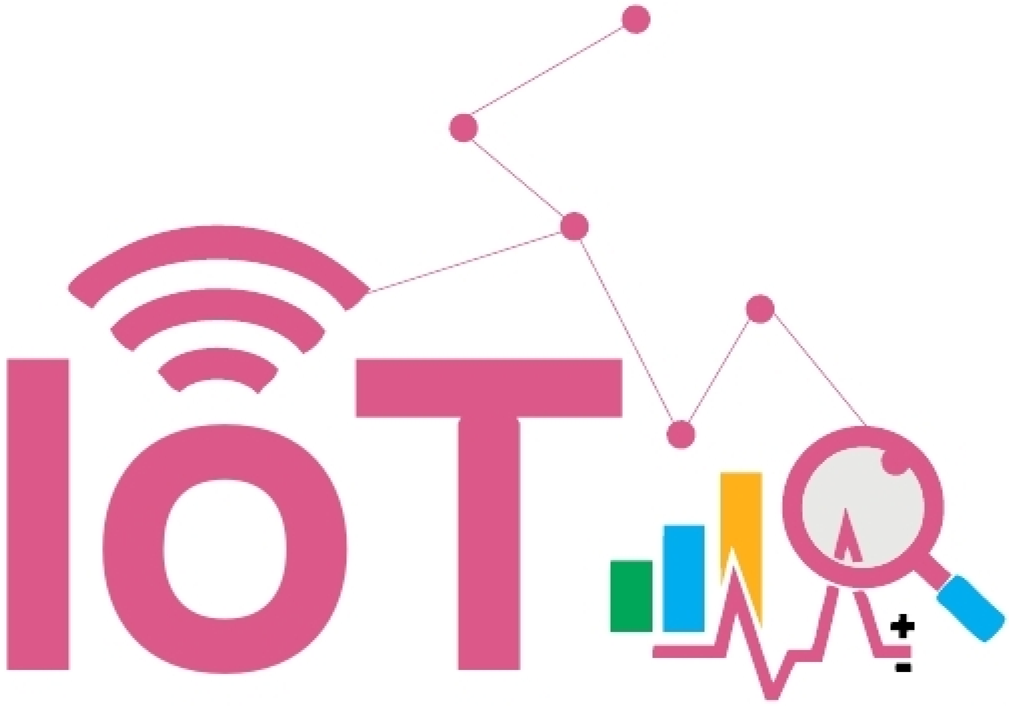

Creating New Avenues for Data Analytics

An IoT solution is incomplete without in-built data analytics as a part of the user application layers. Customers look for an effortless way to know the insights from data collected by the IoT hardware using sensors. The greatest value lies in how businesses apply IoT, learn from the data captured, and improve efficiencies.

The Internet of Things (IoT) is a network of 'things' with in-built intelligence to sense, store and exchange information with mobile phones, machines, smart homes, vehicles, electronic devices, etc. It is a part of the fourth industrial revolution, which is basically an evolution of mobile and embedded technologies with connectivity to the Internet, integrating greater computing and analytic capabilities. International Data Corporation (IDC) says, "There will be 41.6 billion connected devices, generating 79.4 zettabytes (ZB) of data by 2025."

With billions of devices connected to the Internet and so many people exchanging information with devices, these can become an intelligent network of systems. These systems, when sharing data over the cloud, can transform products, businesses, and lives in countless ways—whether it's fast medical service, better education, lower transportation costs, or optimising energy generation and consumption.

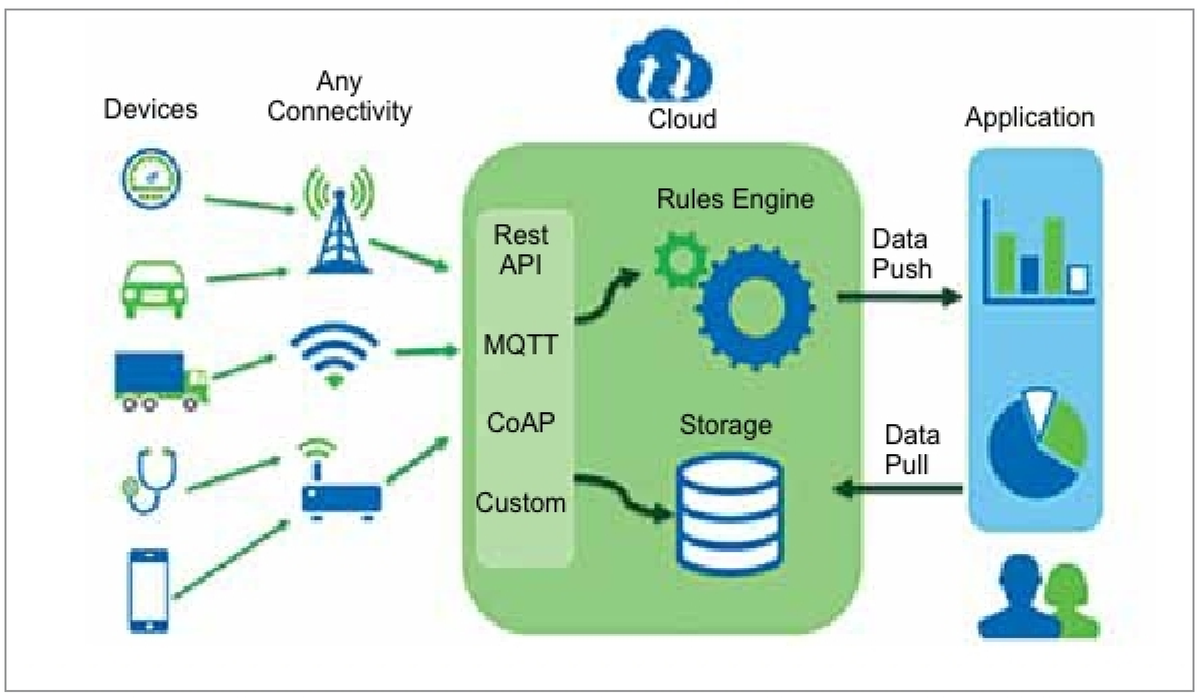

Figure 1: Data analytics application analyses complex data sets generated by different IoT devices and stored in a cloud over Wi-Fi, GSM and Bluetooth connectivity (Credit: *https://itviconsultants.com*)

IoT technology stack

The IoT technology stack contains five different layers, including hardware, device software, communication, cloud, and user applications. The whole stack is required to work in sync to benefit from the IoT technology.

The hardware layer contains the microcontroller, necessary sensors, and peripherals suitable for the product. The device software layer guides the hardware to function as per specifications. It is basically a firmware enabling the hardware. The communication layer implements Internet connectivity using wireless protocols like Bluetooth, Wi-Fi, or GSM based on the range, bandwidth,